

hpsatviews: High-performance command-line tool for geostationary satellite imagery

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Software

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Summary

hpsatviews (hpsv) is a command-line tool that turns raw geostationary weather-satellite data into ready-to-view images. Pointed at a single NetCDF file from a GOES-R series satellite's Advanced Baseline Imager (ABI), it automatically locates the sibling spectral channels it needs, applies atmospheric and contrast corrections, and writes a grayscale, pseudocolor, or RGB composite — a natural “true color” view, a volcanic-ash or air-mass product, or a day/night blend with city lights — as a PNG or a georeferenced Cloud-Optimized GeoTIFF. Following the terminology of Lira Chávez (2010), it treats the source data as an *image*, derives from it a normalized, humanly interpretable *view*, and labels that view as a *product* (true color, ash, air mass, ...) once it matches a concept recognized by the environmental sciences community. Written in C11 and parallelized with OpenMP, hpsv targets near-real-time throughput on commodity hardware: it generates visual products for human interpretation, and is scoped deliberately narrowly to do that one task quickly.

Statement of need

Operational hazard monitoring needs visual satellite products within seconds of data arrival, repeated continuously as new scenes arrive every five to fifteen minutes: a volcanic-ash advisory or a hurricane day/night composite that takes minutes to render has already lost part of its useful window by the time it reaches a forecaster. hpsv's output products are distributed operationally, via FTP and web channels, to Mexican agencies responsible for hazard early warning — including CENAPRED (Centro Nacional de Prevención de Desastres) and SEMAR (Secretaría de Marina) — for which a short, predictable turnaround from satellite overpass to delivered image is a hard operational requirement, not a convenience. The established tools for generating these products, satpy (Pytroll) and geo2grid/Polar2Grid (SSEC/CIMSS) (Space Science and Engineering Center (SSEC), University of Wisconsin-Madison, 2026), are Python libraries built for the breadth of operations a satellite-data researcher might need: dozens of instrument readers, resampling backends, and an interactive analysis workflow, at the cost of a heavier runtime stack. hpsatviews targets the narrower, latency-sensitive niche next to them: a single compiled binary with no Python runtime or external lookup-table files to load, OpenMP-parallel pixel loops, and Rayleigh-correction tables embedded at compile time, producing the same well-established GOES-R products through one CLI invocation per scene. It is aimed at operational satellite-monitoring groups and image-pipeline integrators who need a fast, scriptable, dependency-light component for a larger distribution system, rather than at researchers exploring new algorithms, who are already well served by satpy/geo2grid's flexibility.

40 State of the field

41 By design, hpsatviews does not aim to replace general-purpose GIS platforms such as
42 QGIS/GDAL or full satellite-data analysis libraries such as satpy; it is scoped exclusively
43 to the domain of generating standard views for human interpretation, as fast as possible.
44 Within that narrower scope it deliberately reuses the published methodology of the Python-
45 based geostationary-imaging ecosystem rather than reinventing it: its true-color green-channel
46 synthesis matches the coefficients used by geo2grid/satpy for GOES-R ABI, which lacks
47 a native green band (Bah et al., 2018; Miller et al., 2016); its piecewise contrast stretch
48 and ratio-sharpening ("SelfSharpenedRGB"-equivalent) reproduce geo2grid's; and its default
49 Rayleigh-correction lookup tables are generated from pyspectral (Dybbroe et al., 2018;
50 Scheirer et al., 2018). What hpsatviews changes is the implementation substrate: a self-
51 contained C11/OpenMP binary instead of an xarray/dask-based Python stack, trading satpy's
52 breadth (a broad instrument catalogue, a general analysis API) for a single family of satellites
53 and a single use case. Desktop tools such as QGIS or McIDAS-V cover overlapping visualization
54 ground but are interactive applications, not headless pipeline components. To the author's
55 knowledge, no other actively maintained tool occupies this specific niche: a native, OpenMP-
56 parallel CLI that reproduces the standard GOES-R RGB product catalogue with no runtime
57 dependency beyond the C libraries it links against.

58 Software design

59 hpsatviews is written in strict modern C (C11 with POSIX extensions, built with -Wall
60 -Wextra) and relies on OpenMP parallelization and a small set of efficient design patterns
61 rather than abstraction layers. The processing pipeline is linear and explicit (see Figure 1):
62 parse CLI arguments into an immutable ProcessConfig; load the anchor NetCDF file and
63 infer/load sibling channels based on the requested command mode (single-channel or RGB
64 composite); sequentially apply optional corrections (Rayleigh atmospheric scattering and
65 CLAHE contrast enhancement); normalize to 8-bit; optionally reproject from the satellite's
66 fixed grid to a geographic lat/lon grid; write the final image as PNG or GeoTIFF (where
67 essential metadata is automatically embedded within the GeoTIFF tags for seamless extraction
68 by downstream cataloging tools) and, conditionally, write a JSON metadata sidecar. RGB
69 products are implemented as a mode table mapping each named product (truecolor, ash,
70 airmass, daynite, severestorm, so2, ...) to a per-channel linear combination, plus a custom
71 mode that accepts arbitrary band-algebra expressions (e.g. "2*C13-C14") so users are not
72 limited to the built-in catalogue. Two Rayleigh-correction implementations are offered: a
73 default table lookup derived from pyspectral (Dybbroe et al., 2018; Scheirer et al., 2018) and
74 embedded directly in the binary to avoid any runtime data dependency, and a lighter analytic
75 alternative based on the Bucholtz scattering model and the Hansen and Travis phase function
76 (Bucholtz, 1995; Hansen & Travis, 1974), with both following the Rayleigh optical-depth
77 formulation of Bodhaine et al. (1999) and applying cloud relaxation above a reflectance
78 threshold. Contrast enhancement includes Contrast Limited Adaptive Histogram Equalization
79 following Pizer et al. (1987) and Zuiderveld (1994). The central trade-off throughout is startup
80 latency and dependency footprint versus runtime flexibility: embedding the Rayleigh LUTs
81 at compile time means a rebuild is required to change atmospheric models, but it removes
82 any need for Python, pyspectral, or external data files at run time; likewise, OpenMP loop
83 parallelism trades multi-node scalability for simplicity on the single multi-core machines typical
84 of an image-ingestion pipeline.

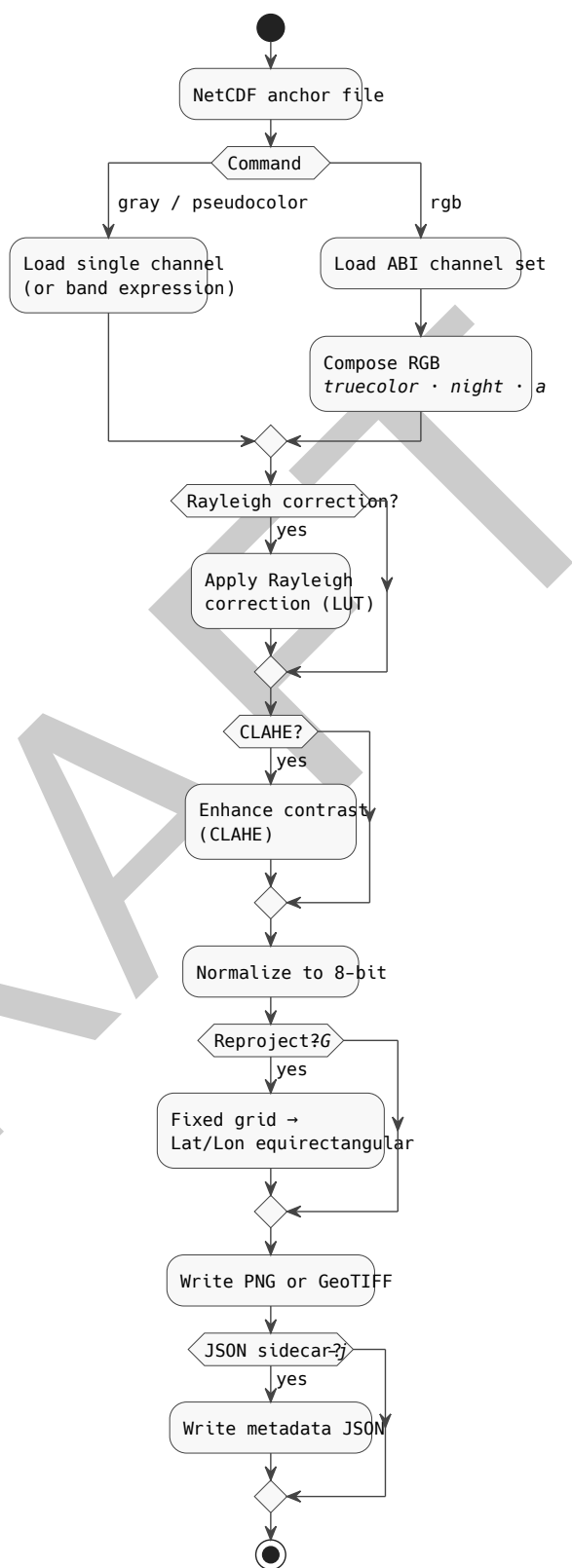


Figure 1: Core processing workflow of hpsatviews. The execution path diverges based on the requested command, followed by sequential, optional stages for Rayleigh atmospheric correction, CLAHE contrast enhancement, and geographic reprojection before writing the final image output and an optional JSON metadata sidecar.

Research impact statement

This operational deployment to CENAPRED and SEMAR, introduced above, is not incidental: it depends on the low-latency, dependency-light design described in the preceding sections. A single pipeline execution can produce both the native full-disk view and a geographically reprojected grid from which operationally relevant regional crops are cut, as illustrated in Figure 2.

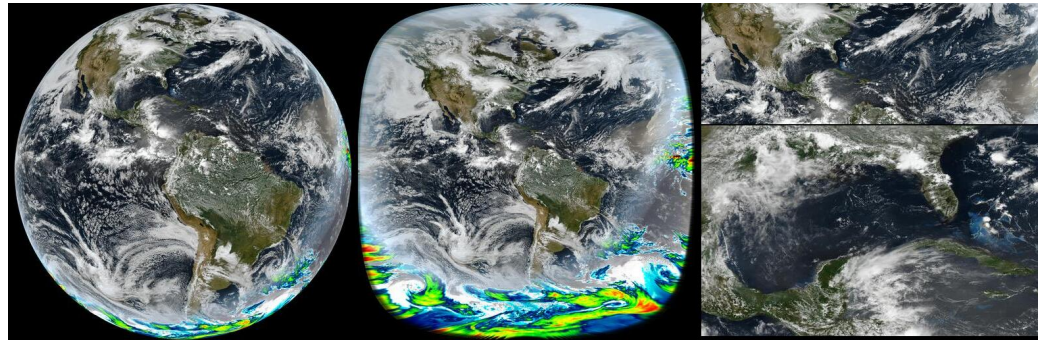


Figure 2: Geometric processing workflow in hpsatviews: (Left) A true color RGB composite generated in the native geostationary projection of the GOES-R Advanced Baseline Imager, showcasing atmospheric Rayleigh correction; (Center) The same full disk computationally reprojected into a uniform geographic latitude/longitude grid (WGS84); (Right) Arbitrary regional crops extracted from the reprojected domain, demonstrating the tool's ability to generate localized, operationally relevant views (e.g., North Atlantic and the Gulf of Mexico) from a single execution pipeline.

It reached its first public release (v1.0.0) in June 2026, archived on Zenodo under a persistent DOI, and is also a component of a larger imagery pipeline at LANOT/UNAM: its optional JSON/GeoTIFF metadata sidecar is designed for, and consumed by, a companion automation tool [mapdrawer](#), that catalogues and serves generated images using the embedded coordinate-reference-system, bounding-box, and product fields. As a first public release under its current name, it does not yet have independent external citations in the literature; its demonstrated near-term significance is this existing operational role in Mexican disaster-prevention and maritime early-warning distribution, plus its general fitness — given its narrow scope and minimal dependency footprint — as a fast component for similar operational infrastructure elsewhere, as demonstrated by the representative sample of application-specific composites shown in Figure 3 (among the many products it supports), in addition to the localized regional crops described above.

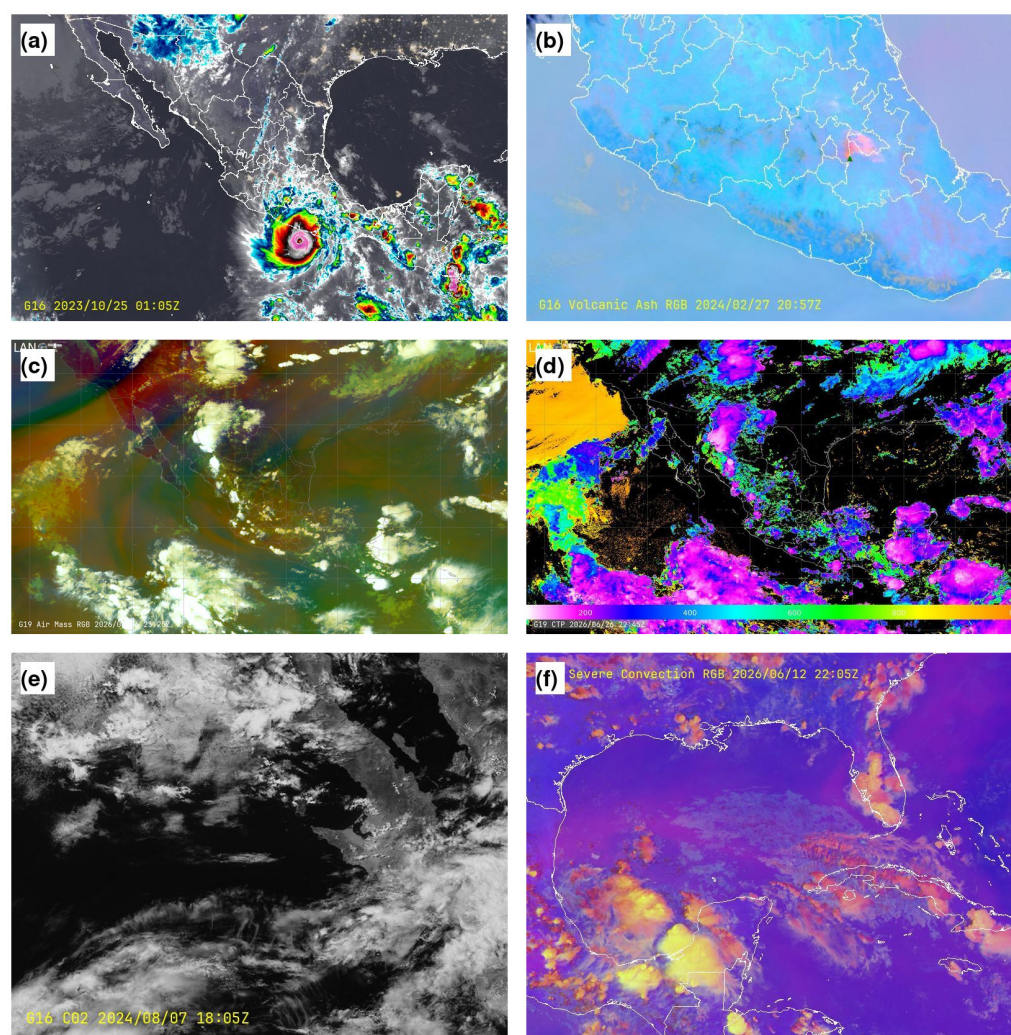


Figure 3: Composite panel of hpsatviews radiometric outputs generated from GOES ABI data: (a) Night composite with meteorological colormap; (b) Volcanic-ash RGB composite isolating SO₂ and ash plumes; (c) Air mass RGB composite illustrating synoptic-scale dynamics; (d) Pseudocolor enhancement of a Level-2 Cloud Top Pressure product; (e) Grayscale view of the high-resolution visible channel with Contrast Limited Adaptive Histogram Equalization (CLAHE) applied to enhance local cloud textures; (f) Severe Storms RGB composite highlighting deep convection.

Future development for hpsatviews will focus on expanding support to additional satellite platforms beyond the GOES-R series. Additionally, exploring fine-grained GPU parallelism is under consideration for the most computationally intensive stages—specifically Rayleigh atmospheric correction and geometric reprojection.

AI usage disclosure

The initial design and overall philosophy of this project are entirely human. As the project evolved, however, large-scale refactoring, the diagnosis of elusive bugs, and the implementation of intricate algorithms benefited substantially from AI assistance — primarily Gemini Pro (versions 2.4 to 3.1) and Claude Sonnet (versions 4.5 to 4.6). AI was also used to draft portions of this paper. In all cases, the human author reviewed, edited, and validated every AI-assisted output and retained sole responsibility for all core design decisions.

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